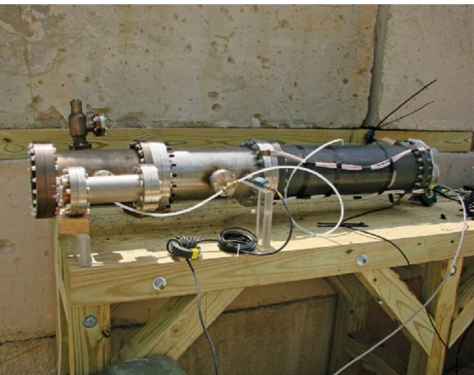


MICROWAVE TUBES MAKING A COMEBACK

Scientists are looking back at the microwave tubes for high-power and high-frequency applications because only these can handle a power of up to 300 megawatts at a frequency of 1 GHz



A high-power microwave system consisting of a high-power microwave tube, high-voltage measurement chamber and power conditioning unit

■ **PRABHAT K. MISHRA, SAHAJ SAXENA**

In 1904, J.A. Fleming introduced the vacuum tube diode. After the second world war, electron tubes were used to develop the first generation of computers but these computers were impractical due to the large sizes of the electronic components. In 1947, John Bardeen, Walter Brattain and William Shockley demonstrated the amplifying action of

the first transistor at Bell Telephone Laboratories. They received a *Nobel Prize* for it.

Bipolar transistors and digital integrated circuits (ICs) were made first. Analogue ICs, large-scale integration (LSI) and very-large-scale integration (VLSI) followed by the mid-1970s. A VLSI design consists of thousands of circuits on a single chip with transistors acting as on/off switches or gates between them. Transistors are good for low-power

and low-frequency applications. Microcomputers, medical equipment, video cameras and communication satellites are all examples of devices made possible by using ICs.

From the day of invention of vacuum tube till today, when millions of transistors are fabricated on a single chip, technology has advanced a lot. But scientists are looking back at the microwave tubes for high-power and high-frequency applications because only these can handle high power (nearly 300 megawatts) at high frequency (nearly 1 GHz).

The term 'microwave' denotes the techniques and concepts used as well as a range of frequencies. Microwaves travel in matter in the same manner as light waves. These are reflected by metals, absorbed by some dielectric materials and transmitted through other materials without significant losses.

Many R&D centres in India are actively involved in the advancement of microwave tubes. These include:

1. Central Electronics Engineering Research Institute, Pilani
2. Central Scientific Instruments Organisation, Chandigarh
3. Central Glass and Ceramic Research Institute, Kolkata
4. Central Mechanical Engineering Research Institute, Durgapur
5. Bharat Electronics Ltd, Bengaluru
6. Vacuum Electronics Devices and Application Society
7. Microwave Tube Research and Development Center, Bengaluru